SECTION 16461-A

DRY TYPE TRANSFORMERS (600 V AND LESS)

PART 1 GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following types of dry-type transformers rated 600 V and less, with capacities up to 1000 kVA:

1). Distribution transformers.

2). Buck-boost transformers.

1.3 SUBMITTALS

A. Product Data Include rated nameplate data, capacities, weights, dimensions, minimum clearances and performance for each type and size of transformer indicated.

B. Shop Drawings: Wiring and connection diagrams.

C. Factory test reports.

D. Output Settings Reports: Record of tap adjustments specified in Part 3.

1.4 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction and marked for intended use.

B. Comply with IEEE C 57.12.91.

C. Transformers Rated 15 kVA and Larger: Certified as meeting NEMA TP 1, Class 1 efficiency levels when tested according to NEMA TP 2.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Temporary Heating: Apply temporary heat according to manufacturer's written instructions within the enclosure of each ventilated-type unit, throughout periods during which equipment is not energized and when transformer is not in a space that is continuously under normal control of temperature and humidity.

1.6 COORDINATION

A. Coordinate size and location of concrete bases.

B. Coordinate installation of wall-mounting and structure hanging supports.

PART 2 PRODUCTS

2.1 MANUFACTURERS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1). ABB

2). SIEMENS

3). REX

2.2 MATERIALS

A. Description: Factory-assembled and -tested, air-cooled units for 60-Hz service.

B. Cores: Grain-oriented, non-aging silicon steel with high magnetic permeability, and low hysteresis and eddy current losses. The core flux density shall be below saturation point to prevent core overheating caused by harmonics even with 10% primary over-voltage.

C. Coils: Continuous windings without splices, except for taps. Impregnated with non-hygroscopic, thermo-setting varnish.

1). Internal Coil Connections: Brazed or pressure type.

2). Coil Material: Copper.

D. Completed core and coil shall be bolted to enclosure base and isolated from base by rubber vibration absorbing mounts.

2.3 DISTRIBUTION TRANSFORMERS

A. Comply with NEMA ST 20, ANSI C89, and list and label as complying with UL 1561.

B. Cores: One leg per phase.

C. Enclosure: Ventilated, NEMA 250, Type 2 (indoor with some degree of protection from falling dirt, dripping, and splashing liquids).

   1). All covers shall have lifting handles.

   2). Wiring compartment shall be suitable for conduit entry and large enough to allow convenient wiring.

D. Indoor Transformer Enclosure Finish: Comply with NEMA 250 for "Indoor Corrosion Protection."

   1). Finish Color: ANSI 61 gray.

E. Insulation Class: 220 deg C, UL-component-recognized insulation system with a maximum of 150 deg C rise above 40 deg C ambient temperature.

   1). Transformer shall be capable of operating at 100% of nameplate rating continuously while in an ambient temperature not exceeding 40 deg C.

F. Electrical ratings: Unless otherwise shown on the drawings, the electrical ratings shall be:

   1). Primary: 480V, 3-phase, delta

   2). Secondary:208Y/120 volts (480Y/277 volts)

   3 - phase grounded

   3). Frequency: 60Hz

   4). KVA rating: As shown on drawings

G. Taps for Transformers:

   1). 3 kVA and Smaller: No taps

   2). Between 3kVA and 25 kVA: One 5% tap above and one 5% tap below normal full capacity.

3). 25 kVA and Larger: Two 2.5% taps above and four 2.5% taps below normal full capacity.

H. K-Factor Rating (for tools equipment only): Unless otherwise shown on the drawings, all 30kVA and larger transformers shall be K-13 rated and comply with UL 1561 requirements for non-sinusoidal load current-handling capability to the degree defined by designated K-factor.

1). Unit shall not overheat when carrying full-load current with harmonic distortion corresponding to designated K-factor.

2). Indicate value of K-factor on transformer nameplate.

I. Electrostatic Shielding: All K-factor rated transformers shall have electrostatic shielding. Each winding shall have an independent, single, full-width copper electrostatic shield arranged to minimize inter-winding capacitance.

1). Arrange coil leads and terminal strips to minimize capacitive coupling between input and output terminals.

2). Include special terminal for grounding the shield.

3). Shield Effectiveness:

a). Capacitance between Primary and Secondary Windings: Not to exceed 33 picofarads over a frequency range of 20 Hz to 1 MHz.
b). Common-Mode Noise Attenuation: Minus 120 dBA minimum at 0.5 to 1.5 kHz; minus 65 dBA mini-mum at 1.5 to 100 kHz.
c). Normal-Mode Noise Attenuation: Minus 52 dBA minimum at 1.5 to 10 kHz.

J. Wall Brackets: Where transformers are shown wall mounted, provide manufacturer's standard brackets.

K. Sound-Level Requirements: the transformer self-cooled ratings shall not exceed NEMA ST 20 and ANSI C89.2 standard sound levels.

1). 9 kVA and Less: 40dBA.

2). 30 to 50 kVA: 45dBA.

3). 51 to 150 kVA: 50dBA.

4). 151 to 300 kVA: 55dBA.

5). 301 to 500 kVA: 60dBA.

6). 501 to 700 kVA: 62dBA.

7). 701 to 1000 kVA: 64dBA.

L. Terminations:

1). All field wire terminations shall be accessible from the front.

2). The secondary neutral terminal shall be twice the ampacity of the secondary phase terminal.

3). Bus bar for cable terminations will be large enough and provided with provisions to allow termination of the number and size of conductors shown on the drawings.

2.4 BUCK-BOOST TRANSFORMERS

A. Description: Self-cooled, two-winding dry type, rated for continuous duty and with wiring terminals suit-able for connection as autotransformer. Transformers shall comply with NEMA ST 1 and shall be listed and labeled as complying with UL 1561.

B. Enclosure: Ventilated, NEMA 250, Type 2.

1). Finish Color: ANSI 61 gray.

2.5 FACTORY TESTS

A. Test and inspect transformers according to IEEE C57.12.91.

B. All transformers shall have the following production tests:

1). Applied Potential

2). Induced Potential

3). No Load Losses

4). Voltage Ratio

5). Polarity

6). Continuity

C. Manufacturer shall perform the following additional tests on units identical to each type of transformer being supplied for this project.

1). Sound Levels

2). Temperature Rise ests

3). Full-Load Losses

4). Regulation

5). Impedance

PART 3 EXECUTION

3.1 EXAMINATION

A. Examine conditions for compliance with enclosure and ambient temperature requirements for each transformer.

B. Verify that field measurements are as needed to maintain working clearances required by NFPA 70 and manufacturer's written instructions.

1). Examine walls and floors for suitable mounting conditions where transformers will be installed.

2). Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install wall mounting transformers level and plumb with wall brackets fabricated by transformer manufacturer. Brace according to manufacturer's written instructions and seismic codes at Project.

B. Install floor-mounted transformers level on concrete bases according to Division 16 Section “Basic Electrical Materials and Methods”.

1). Anchor transformers to concrete bases according to manufacturer's written instructions and seismic codes at Project.

2). Provide 1" thick high resiliency pads to isolate transformer from floor or platform. Korfund "Elasto Rib" or equivalent.

C. Assure that conductors are protected from abrasion from the bottom grating in the transformer.

3.3 CONNECTIONS

A. Ground equipment according to Division 16 Section "Grounding and Bonding."

B. Connect wiring according to Division 16 Section "Conductors and Cables."

C. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A and UL 486B.

D. Connections between panelboards and transformers of 12” or less: Use flexible wire-way fittings, Hoffman or equal.

E. Connections between panelboards and transformers over 12”: Use flexible metal conduit.

3.4 IDENTIFICATION

A. Label each transformer according to Division 16 Section "Basic Electrical Materials and Methods".

3.5 ADJUSTING

A. Set the secondary taps such that the secondary voltage will be as close as possible, but not less than 214 volts at no load.

B. Adjust buck-boost transformers to provide nameplate voltage of equipment being served, plus or minus 5 percent, at secondary terminals.

3.6 Output Settings Report: Prepare a written report recording output voltages and tap settings.

3.7 TESTING

A. Perform visual and mechanical inspections listed in NETA ATS 7.2.1.1 (Transformers, dry type, air-cooled, 600V and below – small).

END OF SECTION